Synthesis: The inverse Fourier transform, calculating the time domain from the frequency domain. See *analysis* for comparison.

System: Any process that produces an output signal in response to an input signal.

Systematic error: Errors in a measurement or prediction that are repeatable from trial to trial. Systematic errors determines *accuracy*. See *random error* for comparison.

Target detection: Deciding if an object or condition is present based on measured values.

TIFF: A common image file format used in word processing and similar programs. Usually not compressed, although LZW compression is an option. See *GIF* and *JPEG* for comparison.

Timbre: The human perception of harmonics in sound. See *pitch* for comparison.

Time domain: A signal having time as the independent variable. Also used as a general reference to any domain the data is acquired in.

Time domain aliasing: Aliasing occurring in the time domain when an action is taken in the frequency domain. Circular convolution is an example.

Time domain encoding: Signal information contained in the shape of the waveform. See *frequency domain encoding* for comparison.

Transfer function: The output signal divided by the input signal. This comes in several different forms, depending on how the signals are represented. For instance, in the s-domain and z-domain, this will be one polynomial divided by another polynomial, and can be expressed as *poles* and *zeros*.

Transform: A procedure, equation or algorithm that changes one group of data into another group of data.

Transform compression: Data compression technique based on assigning fewer bits to the high frequencies. *JPEG* is the best example.

Transition band: Filter jargon; the band of frequencies between the passband and stopband where the roll-off occurs.

True-negative: One of four possible outcomes of a target detection trial. The target is not present, and is correctly indicated to be not present.

True-positive: One of four possible outcomes of a target detection trial. The target is present, and correctly indicated to be present.

Unit circle: The circle in the z-plane at $r = 1$. The values along this circle are the frequency response of the system.

Unit impulse: Another name for *delta function*.

Von Neumann Architecture: Internal computer layout where both the program and data reside in a single memory; very common. See *Harvard Architecture* for comparison.

Voiced: Human speech sound that originates as pulses of air passing the vocal cords. Vowels are an example of voiced sounds. See *fricative* for comparison.

Well: Short for *potential well*; the region in a CCD that is sensitive to light.

White noise: Random noise that has a flat frequency spectrum. Occurs when each sample in the time domain contains no information about the other samples. See *1/f noise* for comparison.

Wiener filter: Optimal filter for increasing the signal-to-noise ratio based on the frequency spectra of the signal and noise.

Windowed-sinc: Digital filter used to separate one band of frequencies from another.

z-domain: The domain defined by the z-transform. Also called the *z-plane*.

z-transform: Mathematical method used to analyze discrete systems that are controlled by difference equations, such as recursive (IIR) filters. Changes a signal in the time domain into a signal in the z-domain.

Zero: A term used in the Laplace & z-transforms. When the s-domain or z-domain transfer function is written as one polynomial divided by another polynomial, the roots of the numerator are the *zeros* of the system. See also *pole*.

Zero phase: A system with a phase that is entirely zero. Occurs only when the impulse response has left-to-right symmetry around the origin. See also *linear phase*.

Zeroth-order hold: A term used in DAC to describe that the analog signal is maintained at a constant value between conversions, resulting in a staircase appearance.